

TemporalTemp: Encoding Historical Temperatures for Automated Sea Ice Mapping

Jayden Hsiao*
Systems Design Engineering
University of Waterloo
Email: j3hsiao@uwaterloo.ca

David A. Clausi
Systems Design Engineering
University of Waterloo
Email: dclausi@uwaterloo.ca

Lincoln Linlin Xu
Geomatics Engineering
University of Calgary
Email: linlin.xu@ucalgary.ca

Abstract—This paper investigates the impact of incorporating historical ERA5 reanalysis data into deep learning models for Arctic sea ice parameter prediction. While existing approaches primarily rely on contemporaneous observations, we explore whether encoding temporal variations in ERA5 2-meter air temperature (t2m) can improve classification of sea ice concentration (SIC), stage of development (SOD), and floe size (FLOE). Using a multi-task U-Net architecture, we evaluate different temporal encoding strategies on the AI4Arctic dataset, comparing performance against a baseline model without historical context. Our results show that a 1-year encoding with 4 samples of 90 days each achieves the best overall performance (85.62% combined score), improving SOD and FLOE classification by 1.43% and 1.20% respectively. Qualitative analysis reveals that temperature encoding helps distinguish ice types during transitional seasons but can degrade SIC estimation during melt conditions when air temperatures become misleading indicators. These findings demonstrate that carefully designed temporal encodings can enhance sea ice classification while highlighting the need for complementary data sources to address limitations during melt periods.

I. INTRODUCTION

Sea ice plays a crucial role in global climate regulation, maritime safety, and the livelihoods of indigenous communities that rely on stable ice conditions for travel and hunting [1]–[3]. However, Arctic ice conditions are becoming increasingly variable, with trends such as delayed freeze-up, increased fragmentation, and reduced summer ice extent [4]. These changes pose challenges for navigation, ecosystem stability, and traditional ways of life, emphasizing the need for continuous monitoring.

Traditionally, sea ice conditions are assessed through manually produced ice charts, which rely on expert interpretation of remote sensing data within geographic information system software. While these charts provide invaluable insights, their production is labor-intensive and limited in both spatial and temporal coverage, underscoring the need for automated approaches [5].

Key indicators such as sea ice concentration (SIC), stage of development (SOD), and floe size (FLOE) are widely used to characterize Arctic ice dynamics. SIC measures the fraction of ice-covered water, informing climate models and navigation. SOD provides insight into ice age and thickness, aiding in route planning for ice-capable vessels. FLOE describes ice fragmentation, influencing drift patterns and melt processes

[5]. Recent advances in machine learning, particularly deep learning architectures such as U-Nets, have shown promise in automating the estimation of these indicators using synthetic aperture radar (SAR) and passive microwave data [6]–[8]. However, these models primarily rely on contemporaneous observations, limiting their ability to capture longer-term temporal trends in ice evolution.

Arctic change is not only spatial but also strongly influenced by temporal factors, such as melt onset and freeze-up timing. The persistence of specific temperature conditions over time is a key driver of ice formation and decay [9]. While some studies have integrated spatiotemporal encodings to improve model predictions [10], interannual variability remains a challenge. Historical air temperature records have proven effective in capturing key seasonal transitions, such as melt duration and freeze-up timing, making them a promising addition to existing models [11].

This study investigates whether incorporating temporal variations in 2-meter air temperature can improve machine learning models for sea ice parameter prediction. The remainder of this paper is structured as follows: Section II reviews relevant work in sea ice forecasting and deep learning approaches. Section III details our model architecture and temporal encoding strategy. Section IV describes the experimental design and evaluation metrics. Section V presents our findings and analyzes model performance. Finally, Section VI summarizes key insights and discusses directions for future work.

II. BACKGROUND

A. Use of Temporal Data for Sea Ice Forecasting

Several studies have demonstrated the value of incorporating temporal data, particularly ERA5 reanalysis, for sea ice forecasting. Asadi et al. [12] used ERA5 data to develop a sequence-to-sequence deep learning model for probabilistic sea ice forecasts, achieving skillful predictions up to 90 days in advance. Similarly, Kim et al. [13] leveraged ERA5 variables such as temperature, wind speed, and solar radiation to train a U-Net model for long-term SIC forecasting, outperforming traditional statistical and deep learning baselines. Liu et al. [14] employed convolutional long short-term memory (ConvLSTM) networks to predict short-term Arctic sea ice dynamics, highlighting the role of reanalysis climate variables in improving forecast accuracy. These studies underscore the importance

of historical temperature, wind, and radiation trends for sea ice prediction.

B. AutoICE Challenge

The AutoICE Challenge [5] explored the development of deep learning models for multi-parameter sea ice mapping using SAR imagery. As part of this challenge, models were trained and evaluated using the standardized AI4Arctic dataset [15]. The challenge introduced a paradigm shift from single-variable estimation to multi-task learning, evaluating models on their ability to predict SIC, SOD, and FLOE. The winning model, based on a multi-task U-Net architecture [10], achieved state-of-the-art performance, attaining an overall score of 86.39% and an SIC R^2 score of 92.02%. This model demonstrated the effectiveness of integrating multiple data sources, including SAR, passive microwave, and reanalysis data, for automated sea ice characterization.

C. Datasets

This study enhances sea ice mapping by extending the AI4Arctic dataset with historical reanalysis data. While AI4Arctic includes ERA5 climate variables at the time of each scene, we incorporate preceding reanalysis data to provide additional temporal context for sea ice classification.

1) *AI4Arctic Dataset*: The AI4Arctic dataset [15] consists of 533 georeferenced scenes collected between January 2018 and December 2021, covering regions in the Canadian and Greenlandic Arctic. Each scene includes:

- **Sea Ice Charts**: Manually annotated by the Canadian Ice Service (CIS) and Greenland Ice Service (DMI), following the SIGRID-3 standard. These charts are provided as three discrete maps representing SIC, SOD, and FLOE. SIC is reported as the total polygon sea ice concentration, while SOD and FLOE are reported as the ice type associated with the dominant partial concentration. SOD and FLOE polygons are masked where there is no dominant partial concentration (defined as $\geq 65\%$).
- **Synthetic Aperture Radar (SAR) Imagery**: Sentinel-1 C-band SAR images (HH and HV polarizations) acquired in extra-wide swath mode, preprocessed with noise correction and incidence angle adjustment.
- **Passive Microwave Radiometry**: Brightness temperature (TB) measurements from AMSR2, covering multiple frequency bands, resampled to align with Sentinel-1 imagery.
- **Reanalysis Climate Data**: ERA5 climate variables interpolated to match the Sentinel-1 resolution, capturing atmospheric conditions at the time of image acquisition.

The dataset also includes a *ready-to-train* (RTT) version, where SAR images are downsampled to 80m pixel spacing, and normalization is applied across all variables to facilitate deep learning model training.

2) *Historical ERA5 Dataset*: ERA5 is a state-of-the-art climate reanalysis dataset produced by the European Centre for Medium-Range Weather Forecasting (ECMWF) [16]. It provides global estimates of atmospheric, land, and oceanic

climate variables with a spatial resolution of approximately 31km and an hourly temporal frequency, covering the period from 1979 to the present. Observational data are assimilated using a four-dimensional variational (4D-Var) data assimilation scheme, ensuring consistency across time and space.

D. U-Net Architectures for Sea Ice Mapping

U-Net architectures [17] are widely used for segmentation tasks. The model follows an encoder-decoder structure built on convolutional neural networks (CNNs), where downsampling layers extract hierarchical features and upsampling layers restore spatial resolution. Skip connections bridge corresponding encoder and decoder layers, preserving fine-grained details that would otherwise be lost in deeper layers. Each convolutional layer applies a set of filters to learn spatial patterns, expanding the model's receptive field while maintaining local feature representation. In sea ice mapping, U-Nets have demonstrated strong performance in both SIC estimation and classification tasks [6], [7]. Their effectiveness is further highlighted by the AutoICE Challenge, where two of the top five teams employed U-Net-based models [5] to process high-resolution remote sensing data.

III. METHODOLOGY

This study builds upon the architecture and input choice developed in the winning solution of the AI4EO AutoICE Challenge by Chen et al. [10], where the input channels were selected based on their best performance in the AI4Arctic dataset. We extend the model by incorporating historical reanalysis data to enrich the temporal context for sea ice mapping and classification.

TABLE I
MODEL INPUT CHANNELS ADAPTED FROM CHEN ET AL. [10], WITH HISTORICAL ERA5 TEMPERATURE FEATURES (n TEMPORAL SAMPLES) ADDED. SEE SECTION IV-A FOR SAMPLING DETAILS.

Data Type	Feature	Number of Channels
Sentinel-1	Dual-polarized (HH, HV) SAR images	2
	Gridded incidence angle measurements	1
	Distance-to-land map	1
AMSR2	18.7 GHz H, V and 36.5 GHz H, V brightness temperature	4
ERA5	Eastward, northward 10 m wind speed	2
	2 m air temperature	1
	Total column water vapor	1
	Total column cloud liquid water	1
	Historical 2 m air temperature	n
Auxiliary	Gridded latitude, longitude	2
	Acquisition month	1

A. Data Preprocessing and Co-registration

To enhance the temporal context for sea ice mapping, ERA5 2-metre air temperature ($t2m$) data was acquired from

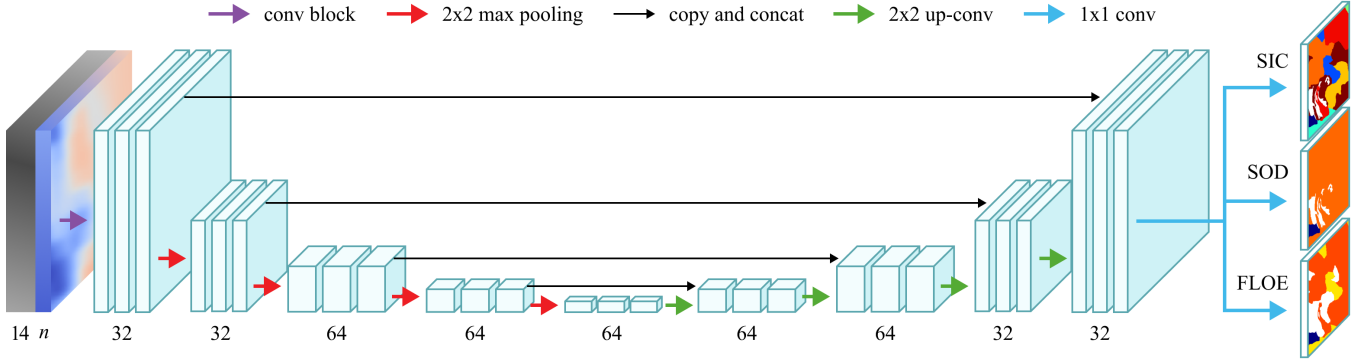


Fig. 1. Diagram of the multi-task U-Net architecture used in all experiments. The labels below each layer represent the number of channels, where n is the number of $t2m$ samples. Encoded historical temperatures are fused at the input-level with the 14 base channels detailed in Table I.

the Copernicus Climate Data Store [18]. The $t2m$ variable represents the air temperature 2 metres above the Earth’s surface, calculated by interpolating between the lowest model level and the Earth’s surface [18].

For this study, historical $t2m$ data from January 2017 to December 2021 were incorporated to complement the AI4Arctic dataset. The chosen range covers the dataset period, as well as an additional year for historical context. The historical $t2m$ data were aligned with the 21×21 latitude and longitude range of the scene, ensuring spatial correspondence. The data were then normalized using the AI4Arctic dataset’s provided mean and standard deviation for $t2m$, ensuring consistent scaling across all time periods. After normalization, the data were resized to match the resolution of the scene using nearest neighbour interpolation, a method chosen for its computational efficiency.

To encode the historical $t2m$ data, we applied a sampling strategy based on time windows parameterized by the number of samples (n) and the duration of each sample window (d , in days). For example, with $n = 12$ and $d = 7$, the encoding captures the past twelve weeks, where each sample represents a contiguous and non-overlapping 7-day period, starting from the scene date and moving backward in time. These encoded time windows were concatenated with the 14 input channels used in the winning AutoICE solution [10], creating a final input to the U-Net model as summarized in Table I.

B. Architecture

The model utilizes a multi-task U-Net architecture designed to simultaneously predict three sea ice parameters: SIC, SOD, and FLOE. As shown in Figure 1, the network consists of four encoder-decoder blocks. The encoder progressively reduces spatial dimensions while capturing high-level features, and the decoder reconstructs these features back to the original input size.

For SOD and FLOE, pixel-wise classification is performed using separate 1×1 convolutional layers at the decoder’s output. Each of these layers corresponds to the number of classes, and the cross-entropy (CE) loss function is applied for classification. On the other hand, SIC is treated as a regression task,

where a dedicated regression head is appended to the final decoder output. This regression layer is optimized using the Mean Squared Error (MSE) loss function.

The architecture is trained end-to-end, with multi-task optimization allowing for joint learning of the three sea ice parameters. This configuration ensures that shared representations across the tasks are learned effectively, while still maintaining task-specific outputs for classification and regression.

IV. EXPERIMENTAL SETUP

A. Temporal Encodings

In this study, we explore various temporal encoding configurations to balance temporal coverage, detail, and data efficiency for the ERA5 2m air temperature ($t2m$) data. As detailed in Section III-A, temporal encodings are defined by two parameters: n , the number of samples, and d , the length of each sample in days.

The temporal coverage of the encodings ranges from a minimum of three months to a maximum of one year. Different combinations of n and d are used to capture various temporal dependencies within these spans. For instance, one year can be represented by $n = 2$ and $d = 180$, which corresponds to two samples of 180 days each. To achieve a finer temporal resolution, $n = 24$ and $d = 15$ also covers one year, but with 24 samples of 15 days each.

These selected time spans align with established work using ERA5 reanalysis data, such as the 90-day forecasting [12] and the 12-month forecasting [13], which have demonstrated the utility of similar temporal resolutions in climate and environmental predictions.

B. Hyperparameters and Training Setup

The hyperparameters for training were adapted from the approach outlined in the AutoICE winning solution [10].

The dual-polarized SAR images, distance maps, and corresponding ice-chart-derived label maps were downsampled by a factor of 10. During training, patches of size 256×256 were randomly extracted from these downsampled images. The AMSR2 and ERA5 variables were resampled to match this size, with their values interpolated within the geographic

TABLE II

SUMMARY OF SCORES FOR DIFFERENT TEMPORAL SAMPLE CONFIGURATIONS ACROSS 20 TEST SCENES. METRICS ARE REPORTED AS THE MEAN SCORE $\pm$ THE STANDARD DEVIATION ACROSS 10 INDEPENDENT CROSS-VALIDATION RUNS. THE BEST PERFORMANCE IN EACH SCORE COLUMN IS HIGHLIGHTED IN **BOLD**.

Encoding Length	t2m Samples n	Sample Window d [Days]	Test Scores [%]			
			Combined	SIC r2	SOD f1	FLOE f1
Baseline (Chen et al. [10])	0	0	85.00 $\pm$ 0.49	90.23 $\pm$ 0.35	86.10 $\pm$ 0.86	72.33 $\pm$ 1.14
3 Months	12	7	85.20 $\pm$ 0.76	89.78 $\pm$ 0.42	86.46 $\pm$ 1.54	73.54 $\pm$ 1.08
	24	3	84.40 $\pm$ 0.86	89.01 $\pm$ 0.54	85.75 $\pm$ 1.37	72.48 $\pm$ 2.53
4 Months	9	14	85.35 $\pm$ 1.06	89.82 $\pm$ 0.64	87.14 $\pm$ 1.62	72.86 $\pm$ 1.49
1 Year	2	180	84.81 $\pm$ 0.69	89.70 $\pm$ 0.55	85.83 $\pm$ 1.52	73.00 $\pm$ 1.08
	3	120	85.39 $\pm$ 1.62	89.61 $\pm$ 1.14	87.20 $\pm$ 2.63	73.31 $\pm$ 2.16
	4	90	85.62 $\pm$ 0.75	89.75 $\pm$ 0.71	87.53 $\pm$ 1.16	73.53 $\pm$ 1.10
	6	60	85.32 $\pm$ 0.94	89.32 $\pm$ 0.73	87.32 $\pm$ 1.52	73.33 $\pm$ 1.06
	12	30	85.14 $\pm$ 0.80	89.36 $\pm$ 0.64	86.95 $\pm$ 1.60	73.08 $\pm$ 1.11
	24	15	84.44 $\pm$ 0.82	88.86 $\pm$ 0.70	86.16 $\pm$ 1.51	72.17 $\pm$ 2.27

areas corresponding to the patches. For validation and testing, entire SAR scenes and distance maps were downsampled, combined with upsampled variables, and input into the trained model, with outputs interpolated back to their original size for evaluation.

The model was trained using Stochastic Gradient Descent (SGD) with momentum, a learning rate of 0.001, and a weight decay coefficient of 0.01. A cosine annealing scheduler [19] was used to adjust the learning rate dynamically, gradually decreasing it until a minimum value before resetting every 20 epochs. This strategy helps the model explore different regions of the loss landscape and avoid suboptimal local minima.

To increase training diversity and improve generalization, data augmentation techniques such as random rotation, flipping, random scaling, and CutMix were applied to SAR images and their corresponding label maps. Training was conducted with a batch size of 64, 128 iterations per epoch, and a total of 100 epochs, selected based on training time constraints and convergence behavior. Since the Mean Squared Error (MSE) loss for SIC is considerably higher than the Cross-Entropy (CE) loss for SOD and FLOE, a loss weighting strategy was applied to balance their contributions, setting the loss weights as $\text{SIC} \times 1$, $\text{SOD} \times 3$, and $\text{FLOE} \times 3$.

All experiments are conducted on the Narval cluster of Compute Canada [20] using an NVIDIA A100-SXM4-40GB GPU and 128 GB of memory with the PyTorch 1.12 library.

C. Evaluation Metrics

Model accuracy was assessed using four key metrics from the AutoICE challenge: the coefficient of determination (R^2) for SIC, the F1 score for SOD and FLOE, and a combined score that weighted these metrics as 2/5 for SIC, 2/5 for SOD, and 1/5 for FLOE [5].

The R^2 score was selected for SIC because it quantifies the variance explained by the model, effectively capturing how closely predictions align with ground truth values. Since SIC is represented as 11 discrete classes corresponding to ice concentration levels in tenths from 0 to 100%, using R^2

ensures that predictions account for the relative proximity between these classes.

For SOD and FLOE classification, we employed the F1 score since these categorical classifications lack consistent quantitative intervals needed for regression metrics. SOD classification spans ice development from new ice (< 10 cm) to multi-year ice (survived > 2 melt seasons), while FLOE sizes range from small floes (< 100 m) to ice fields (> 2 km).

D. Data Splitting and Cross-Validation

To ensure comparability with prior work, the same 20 test scenes from the AutoICE competition were used. For each of the 10 independent runs, we randomly selected 20 validation scenes from the 513 available training scenes for model tuning.

After each epoch, a combined performance score was calculated on the validation set. If the score exceeded the best recorded value from previous epochs, the model parameters were updated and saved. The final model, selected based on peak validation performance, was then used to generate predictions for the test set.

V. RESULTS AND DISCUSSION

Nine configurations of historical temperature encodings were evaluated. Table II summarizes model performance on the 20-scene test set across 10 independent cross-validation runs, where the ($n = 4, d = 90$) encoding achieves the highest combined score of 85.62%. While the Chen et al. [10] baseline achieved superior SIC estimation (90.23%), our temperature-enhanced model demonstrated consistent improvements in both SOD (87.53%, an increase of +1.43% over the baseline) and FLOE classification (73.53%, +1.20% improvement).

Experiments on temporal encoding parameterization reveal two key insights. First, longer encoding windows generally improve model performance by providing a more comprehensive representation of temperature trends over time. However, the way these longer windows are encoded plays a crucial role in determining effectiveness. Encoding with too few samples over long periods (e.g., $n = 2, d = 180$) fails to capture sufficient detail in temperature variations, limiting its usefulness for

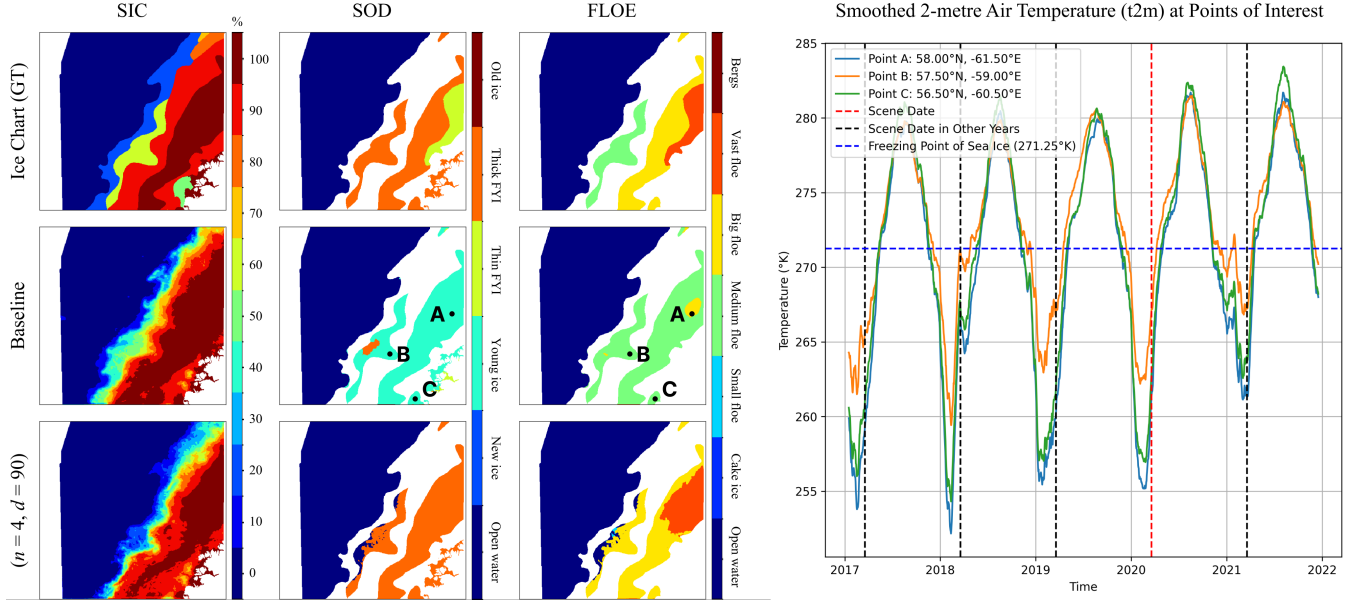


Fig. 2. Qualitative comparison of sea ice classification on March 19, 2020, along the north-eastern coast of Labrador, Canada. The left panel shows sea ice maps across ground truth, baseline, and the best-performing temporal encoding model ($n = 4, d = 90$), highlighting misclassifications by the baseline model at points A, B, and C. The right panel presents 2-metre air temperature trends at these points, averaged over a 30-day rolling window.

prediction. Conversely, excessively high temporal resolutions (e.g., $n = 24, d = 3$ and $n = 24, d = 15$) introduce redundancy, leading to overfitting. In these cases, the model may focus too much on minor fluctuations in temperature rather than prioritizing more informative input channels, such as SAR data.

To better understand the effects of historical temperature encoding on model performance, we analyze two representative test scenes: one where the ($n = 4, d = 90$) model outperforms the baseline, and another where it underperforms. These case studies highlight the strengths and limitations of incorporating historical air temperature trends into sea ice classification.

A. March 19, 2020 – North-Eastern Coast of Labrador, Canada

As illustrated in Figure 2, our model demonstrated improved performance over the baseline in both SOD and FLOE classification for this scene. The left panel of the figure highlights several key misclassifications by the baseline model, where thick first-year ice (FYI) was mistakenly labeled as young ice (SOD), and large floes were misclassified as medium floes at points A, B, and C.

The right panel presents 2-metre air temperature (t_{2m}) trends on March 19 across different years, revealing notable differences in temperature patterns. To highlight these patterns, this temperature is smoothed using a 30-day rolling window. In 2018, temperatures at point B exceeded the freezing point of sea ice (271.25K), whereas in both 2019 and 2020, temperatures remained lower. However, despite similar air temperatures in 2019 and 2020, the melt season in 2020 started slightly later. These variations likely contributed to

our model’s improved ability to classify older and thicker ice types, as it effectively leveraged temperature trends to distinguish ice conditions.

This result underscores the importance of directly encoding temperature rather than relying solely on calendar-based indicators. Since interannual variability affects melt onset and freeze-up timing [11], encoding only the month fails to capture these variations adequately. The same period in different years may exhibit significantly different temperature trends, affecting ice formation and decay. By incorporating historical air temperature records, the model gains a more accurate representation of seasonal transitions. This approach addresses a key challenge in sea ice modeling by allowing the model to adapt to shifting environmental conditions rather than assuming fixed seasonal patterns.

B. July 16, 2018 – North-Eastern Coast of Baffin Island, Canada

Figure 3 presents the results for SIC classification in this scene, where the ($n = 4, d = 90$) temporal encoding model underperformed compared to the baseline. The left panel highlights notable misclassifications made by the temporal encoding model at points D and E. At point D, the model failed to correctly identify consolidated ice regions (100% SIC), misclassifying them as lower SIC categories. Similarly, at point E, areas with 50% SIC were incorrectly labeled as open water, leading to an underestimation of SIC.

The right panel of Figure 3 provides insight into the potential cause of these errors by showing smoothed t_{2m} trends. Despite the presence of consolidated ice, air temperatures at the scene date exceeded the freezing point of sea

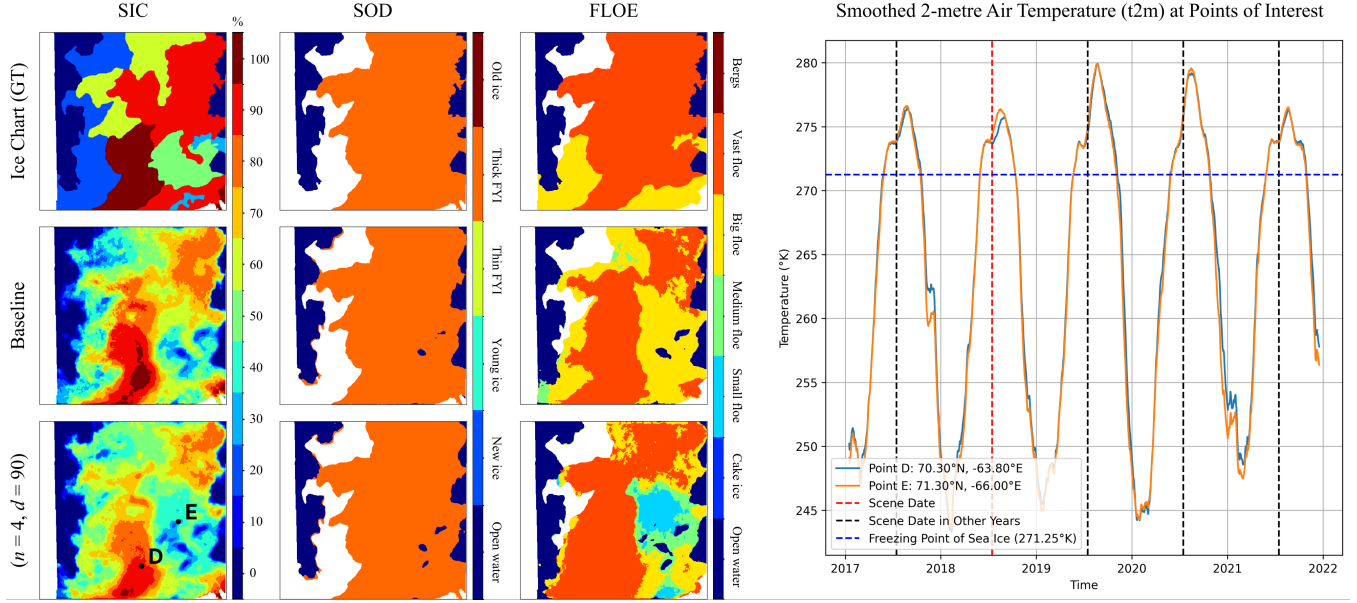


Fig. 3. Qualitative comparison of SIC classification on July 16, 2018, along the north-eastern coast of Baffin Island, Canada. The left panel shows sea ice maps across ground truth, baseline, and the best-performing temporal encoding model ($n = 4, d = 90$), highlighting misclassifications by the temporal encoding model at points D and E. The right panel presents 2-metre air temperature trends at these points, averaged over a 30-day rolling window.

ice, suggesting that surface melt may have decoupled the atmosphere from underlying ice. This effect likely weakened the predictive power of temperature trends, as the temporal encoding model overestimated melting, misinterpreting the warmer temperatures as indicators of more extensive ice loss.

This result highlights a key limitation of temporal encoding when air temperature trends become misleading. Since melt conditions influence $t2m$, the presence of surface melt can elevate air temperatures without necessarily indicating significant ice deterioration. In such cases, directly encoding $t2m$ may degrade performance by reinforcing spurious correlations.

VI. CONCLUSION

This study demonstrates that incorporating historical air temperature data can improve automated sea ice classification, particularly for stage of development and floe size estimation. Our experiments show that a 1-year temporal encoding with four 90-day samples provides optimal performance, achieving an 85.62% combined score. The temperature-enhanced model shows particular strength in transitional seasons where temperature trends help distinguish between ice types, as evidenced by 1.43% and 1.20% gains in SOD and FLOE classification respectively over the baseline.

However, the benefits of temporal encoding are context-dependent. During melt conditions, air temperature trends can become misleading indicators of ice state, sometimes degrading SIC estimation performance. This suggests that while historical temperature data provides valuable supplementary information, it should not replace direct ice observations from SAR and passive microwave sensors.

Future work should explore several promising directions: (1) incorporating VIIRS ice surface temperature (IST) data to complement air temperature measurements, as IST has shown promise for SIC estimation [21]; (2) developing adaptive weighting mechanisms to reduce the influence of temperature features when they conflict with other sensor data; and (3) investigating hybrid approaches that combine learned temporal encodings with physics-based models of ice-temperature relationships. These advances could further improve the robustness of automated sea ice monitoring systems in our changing Arctic.

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